

Production-Ready Azure Linux VM with Terraform

Overview

This project provisions a production-ready Azure Linux Virtual Machine using Terraform with networking, monitoring, security, and automation best practices.

Features

- Custom Virtual Network and Subnet
- Static Public IP
- SSH Key Authentication
- Cloud-init Bootstrap
- Nginx Auto Installation
- Managed Identity
- Azure Monitor Agent
- Log Analytics Workspace
- Data Collection Rule (DCR)
- Remote Terraform Backend

Terraform Workflow

1. terraform init
2. terraform validate
3. terraform plan
4. terraform apply
5. terraform destroy

Monitoring

Azure Monitor Agent and Log Analytics Workspace were integrated successfully. Heartbeat telemetry was verified using Kusto queries.

Verification Query

```
Heartbeat  
| limit 10
```

Project Outcome

This project demonstrates real-world Azure DevOps and Infrastructure as Code practices suitable for portfolio and production learning environments.